

Chapter 1: AI Productivity & Research

Class 5 – Responsible AI Use, Fact Checking & Citation

Learning Objectives

After completing this lesson, students will be able to:

- Understand the importance of responsible AI use.
 - Explain the ethical use of Artificial Intelligence.
 - Identify the limitations of AI-generated information.
 - Verify AI-generated content using reliable sources.
 - Understand fact checking techniques.
 - Learn the importance of citation and giving credit to original sources.
 - Practice responsible digital citizenship while using AI.
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Introduction

Artificial Intelligence (AI) is a powerful tool that helps students learn, research, write, and solve problems. However, AI is not always correct. Sometimes it may provide incorrect, outdated, or incomplete information. Therefore, students should use AI responsibly by checking facts, verifying information, respecting copyright, and giving proper credit to original sources.

Responsible AI use means using AI honestly, ethically, and safely. It helps students become responsible digital citizens and improves the quality of their academic work.

What is Responsible AI Use?

Responsible AI Use means using Artificial Intelligence safely, ethically, honestly, and respectfully while following academic rules and digital safety practices.

Students should use AI to:

- Support learning.
- Improve research.
- Generate ideas.

- Organize information.
- Enhance creativity.

AI should **assist learning**, not replace independent thinking.

Importance of Responsible AI Use

Responsible AI use helps students:

- Improve learning quality.
 - Build critical thinking skills.
 - Avoid misinformation.
 - Protect personal privacy.
 - Respect intellectual property.
 - Develop academic honesty.
 - Become responsible digital citizens.
 - Use technology safely.
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Understanding AI Limitations

AI is powerful, but it has limitations.

AI may:

- Provide incorrect information.
- Generate outdated facts.
- Miss important details.
- Misunderstand questions.
- Create biased responses.
- Produce fictional or inaccurate references.

Students should never assume every AI response is completely accurate.

What is Fact Checking?

Fact Checking is the process of verifying whether information is accurate, reliable, and supported by trusted sources before using it.

Fact checking helps prevent the spread of incorrect information.

Steps for Fact Checking AI Responses

Step 1 – Read the AI Response Carefully

Understand the information before accepting it.

Step 2 – Compare with Reliable Sources

Check information using:

- School textbooks
 - Government websites
 - Educational institutions
 - University websites
 - Scientific journals
 - Official reports
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Step 3 – Compare Multiple Sources

Do not depend on one source.

Compare:

- AI responses
 - Books
 - Educational websites
 - Research articles
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Step 4 – Verify Dates and Statistics

Ensure:

- Data is current.
- Numbers are accurate.
- Statistics come from trusted organizations.

Step 5 – Use Verified Information

Only include information that has been checked and confirmed.

Reliable Information Sources

Students should prefer:

- Government websites
- Educational portals
- Universities
- Research organizations
- Scientific publications
- Libraries
- Textbooks
- Official reports

Reliable sources improve research accuracy.

What is Citation?

A **Citation** is a reference that gives credit to the original author, book, website, article, or research source used in your work.

Citation shows honesty and respects intellectual property.

Why is Citation Important?

Citation helps students:

- Give proper credit.
- Avoid plagiarism.
- Improve academic honesty.
- Support research findings.
- Build credibility.

- Respect copyright.
 - Help readers find original sources.
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Basic Citation Elements

A simple citation usually includes:

- Author Name
 - Title
 - Website or Book Name
 - Publisher (if applicable)
 - Publication Date
 - URL (for online sources)
 - Date Accessed (if required)
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What is Plagiarism?

Plagiarism means copying someone else's work, ideas, images, or writing without giving proper credit.

Examples include:

- Copying AI-generated text without editing.
- Copying content from websites.
- Copying a friend's assignment.
- Using images without permission or acknowledgment.

Plagiarism is unethical and should always be avoided.

Digital Citizenship and AI Ethics

Responsible digital citizens should:

- Use AI honestly.
- Respect others' work.
- Protect personal information.
- Avoid sharing false information.

- Follow school rules.
 - Use technology respectfully.
 - Report inappropriate content.
 - Think critically before sharing information.
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Best Practices for Responsible AI Use

Students should:

- Ask clear and meaningful questions.
 - Verify every important fact.
 - Use AI as a learning assistant.
 - Write answers in their own words.
 - Mention sources when required.
 - Avoid plagiarism.
 - Protect privacy.
 - Review AI-generated work before submission.
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Real-Life Applications

Responsible AI practices are important in:

Education

- Homework
- Assignments
- Research Projects
- Presentations

Business

- Reports
- Market Research
- Customer Analysis

Healthcare

- Medical Research
- Patient Information

Government

- Public Information
- Policy Research

Media

- News Verification
 - Content Creation
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Common Mistakes While Using AI

Avoid:

- Believing every AI response without checking.
 - Copying AI-generated content directly.
 - Ignoring reliable sources.
 - Sharing false information.
 - Using copyrighted content without permission.
 - Not giving proper citations.
 - Asking vague questions.
 - Depending completely on AI.
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Revision

Students should revise:

- Responsible AI Use
 - AI Ethics
 - AI Limitations
 - Fact Checking
 - Reliable Sources
 - Citation
 - Plagiarism
 - Digital Citizenship
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Did You Know?

- AI can generate convincing answers, but they are not always correct.
 - Professional researchers always verify information before publishing.
 - Proper citation improves the credibility of research and assignments.
 - Responsible AI use combines technology with human judgment and ethics.
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Points to Remember

- AI is a helpful assistant, not the final authority.
- Always verify important information.
- Use trusted and reliable sources.
- Give proper credit through citations.
- Never copy content without acknowledgment.